```
  GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <signal.h>

char *memory;

void handle_sigint(int sig)
{
   printf("\nSignal Interrupt Detected.\n");
   free(memory);
   print("Memory released.\n")
   exit(0);
}
int main(int argc,char *argv[])
{
   int mb;
   if (argc !=2)
   {
        printf("Usage: %s <memory_in_MB>\n",argv[0]);
        return 1;
   }

   mb = atoi(argv[1]);

   size_t size = (size_t)mb * 1024 * 1024;
   memory = malloc(size);

   if ( memory ==NULL)
   {
      printf("Memory allocation failed.\n");
      return 1;
   }

   for(int i = 0 ; i < mb * 1024 * 1024; i += 4096)
   {
       memory[i]=1;
   }
   printf("Memory allocated Successfully.\n");
   printf("Press Ctrl+C to release memory and exit.\n");

   signal(SIGINT, handle_sigint);

   while (1)
   {
       sleep(1);
   }
       return 0;
}
```

```
vboxuser@kashik:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:           5.9Gi       1.8Gi       1.8Gi        46Mi       2.6Gi       4.2Gi
Swap:             0B          0B          0B
vboxuser@kashik:~$ ps aux | grep exp6
vboxuser   17301  0.1  1.6 105172 104212 pts/18  S+   21:14   0:00 ./exp6 100
vboxuser   17307  0.0  0.0  18000  2688 pts/5    S+   21:15   0:00 grep --color=auto exp6
vboxuser@kashik:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:           5.9Gi       1.7Gi       1.9Gi        46Mi       2.6Gi       4.3Gi
Swap:             0B          0B          0B
vboxuser@kashik:~$
```

```
vboxuser@kashik:~$ gcc exp6.c -o exp6
vboxuser@kashik:~$ ./exp6 100
Memory allocated Successfully.
Press Ctrl+C to release memory and exit.
^C
Signal Interrupt Detected.
Memory released.
vboxuser@kashik:~$
```